ROMAN TO INTEGER CONVERSION

1. Basic Information

Program Name: Roman.java

Package: recordprog

Language: Java

2. Objective / Aim

To write a Java method that converts a Roman numeral string into its integer value using the subtractive rule for Roman numerals.

3. System Requirements

Hardware: Standard PC/Laptop; minimum 4 GB RAM; minimum 500 MB free storage.

Software: Windows/Linux/macOS; Java Development Kit (JDK); IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code; Command Prompt/Terminal.

4. Algorithm

1. Start the program.

2. Initialize result and previous value to zero.

3. Traverse the Roman numeral string from right to left.

4. Convert the current character to its integer value using a switch statement.

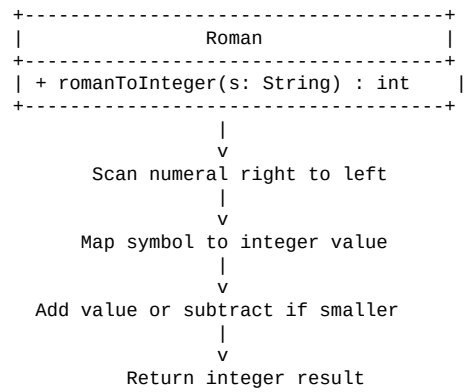
5. If the current value is smaller than the previous value, subtract it from result; otherwise, add it.

6. Store the current value as the previous value.

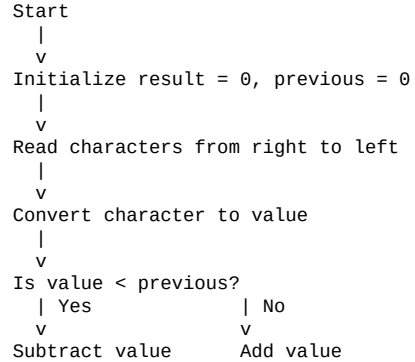
7. Repeat until all characters have been processed.

8. Return the resulting integer and stop.

5. UML / Program Structure



6. Flow of Execution



```
  |                |
  +--------+--------+
           |
           v
Set previous = value
           |
           v
More characters?
  | Yes            | No
  v                v
Repeat loop      Return result
                    |
                    v
                   Stop
```

7. Source Code

```
package recordprog;

public class Roman {
    public int romanToInteger(String s) {
        int result = 0;
        int previous = 0;
        for (int i = s.length() - 1; i >= 0; i--) {
            int value;
            switch (s.charAt(i)) {
                case 'I':
                    value = 1;
                    break;
                case 'V':
                    value = 5;
                    break;
                case 'X':
                    value = 10;
                    break;
                case 'L':
                    value = 50;
                    break;
                case 'C':
                    value = 100;
                    break;
                case 'D':
                    value = 500;
                    break;
                case 'M':
                    value = 1000;
                    break;
                default:
                    throw new IllegalArgumentException(
                        "Invalid Roman numeral character: " + s.charAt(i));
            }
            if (value < previous) {
                result -= value;
            } else {
                result += value;
            }

            previous = value;
        }

        return result;
    }
}
```

8. Compilation and Execution

The provided class contains a conversion method but does not include a main method. To run it as a standalone program, create a main method that instantiates `Roman` and calls `romanToInteger`.

```
javac -d . Roman.java
```

9. Sample Calculation

Input: MCMIV

Calculation: M = 1000, CM = 900, IV = 4

Integer value: 1904

10. Result

Thus, the Java method to convert Roman numeral characters into an integer value was written successfully.